

Multi-Agent Content Generation System – Design Document

1. Overview

This system implements a multi-agent pipeline designed to automate content creation across multiple platforms such as LinkedIn, Twitter/X, Email, and Blog posts. The architecture follows a structured workflow where specialized agents collaborate under the control of an orchestrator. The system emphasizes modularity, clear data flow, and iterative improvement through feedback loops.

2. Agent Roles and Responsibilities

2.1 Orchestrator (Controller Agent)

The orchestrator coordinates the entire workflow and manages communication between agents. It receives user input, triggers agents in sequence, maintains pipeline state, and handles iteration logic such as revision loops.

Input: task description, platforms, brand voice

Output: structured event stream and final results

2.2 Research Agent (Data Collector)

The research agent gathers relevant insights, trends, and audience context related to the task. It provides foundational knowledge used by downstream agents.

Input: task description

Output:

```
{
  "insights": "...",
  "trends": "...",
  "audience_notes": "..."
}
```

2.3 Strategy Agent (Planner)

The strategy agent converts research into a structured messaging framework aligned with the brand voice. It defines tone, messaging, and content angles.

Input: task, research output, brand voice

Output:

```
{
  "core_message": "...",
  "tone": "...",
  "content_angles": ["...", "..."]
}
```

2.4 Content Generator (Executor Agent)

This agent generates platform-specific content using the strategy framework. It adapts tone and format for each selected platform.

Input: strategy framework, platform

Output:

```
{
  "platform": "LinkedIn",
  "content": "Generated post..."
}
```

2.5 Review Agent (Critic / Verifier)

The review agent evaluates generated content for quality, tone, and effectiveness. It determines whether content should be approved or revised.

Input: generated content

Output:

```
{
  "verdict": "APPROVE" | "REVISE",
  "feedback": "Suggestions for improvement"
}
```

2.6 Visual Designer & Optimization Agents

The visual designer generates supporting visual ideas, while the optimization agent enhances content for SEO and engagement.

3. Communication Mechanism

Agents communicate through structured JSON messages managed by the orchestrator. The system streams real-time updates to the frontend using event-based communication.

User → Orchestrator → Research → Strategy → Content → Review → Output

4. Coordination Strategy

The system uses a hybrid model combining a fixed pipeline with an iterative review loop. Content is generated, reviewed, and revised until it meets quality standards or a maximum iteration limit is reached.

Generate → Review → Revise → Repeat (max 2 iterations)

5. Failure and Quality Handling

Low-quality outputs are handled through the review loop. System-level failures such as API errors or timeouts are handled by the orchestrator, which returns error messages and safely terminates the pipeline if needed.

6. Conclusion

This multi-agent system provides a scalable and modular approach to automated content generation. By combining specialized agents with iterative refinement, it ensures high-quality outputs and adaptability for future enhancements.